Middleware Integration Platform Internship Execution Guide

Duration: 4 Weeks (16 Working Days)
Reporting System: Edmapra (Daily)
Presentation Frequency: Every 2 Days

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1. Objective of This Guide

This document defines the exact workflow interns must follow to successfully design, build, and deliver a Middleware Integration Platform.

The goal is to ensure that:

- Work is structured and measurable daily
- Progress is visible through Edmapra reporting
- System development follows real-world engineering practices
- Each phase builds logically toward a complete system

The project itself focuses on building a centralized middleware system that enables secure and controlled communication between multiple applications

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2. Execution Principles

Interns must follow these principles throughout the project:

2.1 Build in Layers

The system must not be built randomly. Each component depends on the previous one:

- Architecture → Gateway → Connectors → Logging → Security → Dashboard

2.2 Daily Accountability (Edmapra)

Each intern must upload a **daily report** containing:

- Tasks completed
- Key technical understanding gained
- Issues encountered
- What will be done next

2.3 Demonstration Over Explanation

Every 2-day presentation must include:

- A working demonstration
- Not just slides or theory

2.4 Incremental Validation

Nothing moves forward unless:

- It runs
- It is tested
- It is understood

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3. Daily Reporting Format (MANDATORY)

Each intern must submit the following on Edmapra every day:

Report Structure

1. Work Done
2. Technical Concepts Understood
3. Challenges Encountered
4. Fixes / Solutions Attempted
5. Next Step (Next Day Plan)

Failure to submit a report = work not done.

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4. Presentation Structure (Every 2 Days)

Each presentation must strictly follow:

1. Objective of the last 2 days
2. What was implemented
3. Live demonstration
4. Technical explanation

5. Issues faced and resolution
6. What comes next

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5. WEEK-BY-WEEK EXECUTION PLAN

WEEK 1: SYSTEM UNDERSTANDING & FOUNDATION

Day 1

Focus: Problem Understanding

Tasks:

- Understand middleware concept
- Identify problems with direct system integration
- List system actors (applications, APIs, users)

Deliverable:

- Written breakdown of system problem
- Initial understanding report

Edmapra Report:

- Must clearly explain “why middleware is needed”

Day 2 (Presentation 1)

Focus: Architecture Design

Tasks:

- Design system architecture
- Identify all components:
    - API Gateway
    - Integration Manager

    - Connectors
    - Logging System
    - Security Layer

Deliverable:

- Architecture diagram

Presentation:

- Explain system flow end-to-end

Day 3

Focus: Project Initialization

Tasks:

- Setup backend project
- Setup Git repository
- Define folder structure

Deliverable:

- Running backend server

Day 4 (Presentation 2)

Focus: Base API Setup

Tasks:

- Create base API (/health)
- Confirm server is running

Deliverable:

- Working backend

Presentation:

- Show project structure and running server

WEEK 2: API GATEWAY & REQUEST FLOW

Day 5

Focus: API Gateway Design

Tasks:

- Design routing logic
- Define how requests enter the system

Deliverable:

- Gateway design notes

Day 6 (Presentation 3)

Focus: API Gateway Implementation

Tasks:

- Implement central route /api
- Add request validation

Deliverable:

- Functional API gateway

Presentation:

- Show request routing

Day 7

Focus: Integration Logic

Tasks:

- Define how system decides where to send requests

Deliverable:

- Decision logic structure

Day 8 (Presentation 4)

Focus: Integration Manager

Tasks:

- Implement routing logic to connectors

Deliverable:

- Working integration decision engine

Presentation:

- Show full request flow

WEEK 3: CONNECTORS & SYSTEM VISIBILITY

Day 9

Focus: Connector Design

Tasks:

- Design connectors:
    - REST API
    - Database

Deliverable:

- Connector structure

Day 10 (Presentation 5)

Focus: Connector Implementation

Tasks:

- Implement at least 2 connectors

Deliverable:

- Working connectors

Presentation:

- Show external communication

Day 11

Focus: Logging System

Tasks:

- Design logging format:
    - Request
    - Response
    - Errors

Deliverable:

- Logging design

Day 12 (Presentation 6)

Focus: Logging Implementation

Tasks:

- Store logs in database

Deliverable:

- Working logging system

Presentation:

- Show logs

WEEK 4: SECURITY, DASHBOARD & FINAL SYSTEM

Day 13

Focus: Security Design

Tasks:

- Define authentication model:
    - API keys or tokens

Deliverable:

- Security design

Day 14 (Presentation 7)

Focus: Security Implementation

Tasks:

- Implement authentication

Deliverable:

- Secured API

Presentation:

- Show protected routes

Day 15

Focus: Monitoring Dashboard

Tasks:

- Build simple dashboard:
    - Requests
    - Success
    - Errors

Deliverable:

- Dashboard UI

Day 16 (Final Presentation)

Focus: Testing & Full Demonstration

Tasks:

- Test full system
- Demonstrate all features

Deliverables:

- Complete working system
- Documentation:
    - Architecture
    - API
    - Deployment

Final Presentation:

- Full system walkthrough
- Real-world scenario demonstration

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6. QUALITY CONTROL EXPECTATIONS

Interns must ensure:

- Code is modular
- Components are reusable
- System is testable
- Logs are meaningful
- Errors are handled

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7. FAILURE CONDITIONS

The project will be considered incomplete if:

- No working API gateway
- No connector integration
- No logging system
- No security enforcement
- No working demonstration

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8. SUCCESS CRITERIA

The project is successful if:

- Multiple systems communicate through the middleware
- All interactions are logged and traceable
- Access is controlled securely
- New integrations can be added easily

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9. FINAL NOTE TO INTERNS

This is not a theoretical exercise.
You are expected to think and operate like engineers building a real system.

Consistency in daily reporting and clarity in presentations will determine the quality of your output.